

PRACTICING WEEK 4

Course: Data Engineering

Goal: Upgrade Practicing 2 into “API → OLTP → Warehouse”.

- I. Add this new section at the start: “API Ingestion (Extract)”.
 - a. Objective is Pull orders from an API and ingest into OLTP tables.
 - b. API endpoints (mock contract)
 - i. GET /api/customers

Response body

```
[
  {
    "id": 1,
    "name": "Alice",
    "email": "alice@example.com"
  },
  {
    "id": 2,
    "name": "Bob",
    "email": "bob@example.com"
  }
]
```

- ii. GET /api/products

Response body

```
[
  {
    "id": 101,
    "name": "Laptop",
    "price": 1200
  },
  {
    "id": 102,
    "name": "Mouse",
    "price": 25
  }
]
```

- iii. GET /api/orders?since=2026-01-01T00:00:00Z&page=1&limit=100

```
[
  {
    "order_id": 1001,
    "customer_id": 1,
    "order_date": "2026-01-02",
    "store": "Store_A",
    "updated_at": "2026-01-02T10:00:00Z",
    "items": [
      {
        "order_item_id": 1,
        "product_id": 101,
        "quantity": 1,
        "price": 1200
      },
      {
        "order_item_id": 2,
        "product_id": 102,
        "quantity": 2,
        "price": 25
      }
    ]
  },
  {
    "order_id": 1002,
    "customer_id": 2,
    "order_date": "2026-01-03",
    "store": "Store_B",
    "updated_at": "2026-01-03T11:00:00Z",
    "items": [
      {
        "order_item_id": 3,
        "product_id": 102,
        "quantity": 1,
        "price": 25
      }
    ]
  }
]
```

iv. GET /api/order-items?order_id=1001

Json Example:

```
{
```

```

"order_id": 1001,
"customer_id": 1,
"order_date": "2026-01-02",
"store": "Store_A",
"updated_at": "2026-01-02T10:00:00Z"
}
[
  {
    "order_item_id": 1,
    "product_id": 101,
    "quantity": 1,
    "price": 1200
  },
  {
    "order_item_id": 2,
    "product_id": 102,
    "quantity": 2,
    "price": 25
  }
]

```

What do you need to apply?

- Deduplicate using primary key (order_id, order_item_id)
- Use watermark (since=updated_at) for incremental loads
- Validate basic quality: not null keys, quantity > 0, price >= 0

II. Create OLTP Database Tables (Storage)

Objective: Prepare the OLTP tables to receive API data.

Create these tables:

- customers(customer_id, name, phone)
- products(product_id, name, category, price)
- orders(order_id, customer_id, order_date, store, updated_at)
- order_items(order_item_id, order_id, product_id, quantity)

```
● ~/MS_CADT_Gen-2/Meng_Oudom-Week_04 main !7 ?12 > python3 create_oltp.py
Connected to PostgreSQL database: de-week3
Tables created successfully.
Existing tables in public schema:
- dim_customer
- dim_date
- dim_product
- fact_sales
- customers
- orders
- order_items
- products
```

What do you need to apply?

- Set PRIMARY KEY for each table
- Add foreign keys for relationships

How to submit?

1. Upload to google drive.